RESCUELOGIC®

A Minimal Event Information Model — Full Technical Working Paper

The SAFER Protocol and the Grammar of a Reported Event

Technical Working Paper — Revision 0.2 | Aligned with proposed SAFER Protocol v1.1

Dan Horon

Cadgraphics Incorporated / RescueLogic

$$e_t = (\text{System}, \text{Identity}, \text{DeviceType}, \text{Location}, \text{Condition}_t)$$

A five-part representation of the meaning of a reported event.

Core event sentence: *This system reports that this device, of this type, at this location, has changed to this state.*

ABSTRACT

Life-safety systems contain large amounts of manufacturer-specific information, yet the information needed to describe a reported point event is remarkably stable. Across different systems, a reported event can be understood through a small set of semantic roles: the reporting system, the identity of the point, the device type, its location, and its reported condition.

Within this representation, Identity, Device Type, and Location describe the relatively stable configuration of an object within a building. Condition is the time-varying property observed during operation, while the reporting System provides source context. These roles are not new facts introduced by SAFER; they make explicit a structure already used implicitly in point lists, drawings, databases, event histories, calculations, and alarm displays.

SAFER also distinguishes the source observation from later classification and response. A point condition can be preserved as reported while receiving systems, operators, or responders apply the classification or action appropriate to their purpose. This separation allows interpretation to be added without rewriting the source facts. In the serialized vocabulary, a compact numeric STATUS code provides a stable machine value, while applications can derive YES, NO, or MAYBE answers to questions without turning uncertainty into a false certainty.

The SAFER information model is independent of serialization and transport. JSON is one way to represent it, and existing communications technologies can carry or map it. This distinction is increasingly useful because modern intelligent systems may retain richer point information than compact legacy reporting formats were designed to convey.

The purpose of SAFER is to provide a small common information layer through which different systems, software applications, and people can understand the basic meaning of a reported event while preserving manufacturer-specific information and the authority of the source system.

Keywords: SAFER • life-safety • event model • semantic interoperability • fire alarm • status normalization • JSON • IoT

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Notation at a glance

e_t	reported event at time t
O	relatively stable object description
C(t)	condition as a function of time
K	derived event classification
R	derived response or action
A(q, e_t, context)	derived YES / NO / MAYBE answer to question q
S, I, T, L, C	System, Identity, Device Type, Location, Condition

1. From Practical Observation to a Formal Model

The model described in this paper can be expressed mathematically, but it developed from observations of real-world fire alarm systems and has been applied across hundreds of installations.

During normal engineering and software work, information from alarm systems repeatedly appeared in groups. The values changed from point to point and from manufacturer to manufacturer, but the groups appeared to contain recurring positions.

ORIGINAL OBSERVATION *One thought was that the values line up like a row of peas in a pod. The peas were different, but the same positions in the pod were consistently filled.*

Those recurring positions became recognizable as different kinds of information:

- which system was reporting;
- which point was involved;
- what kind of point it was;
- where that point was located; and
- what condition it was reporting.

The observation came before the formal terminology. In mathematical and computer-science language, an ordered collection of values can be represented as a tuple. A five-part tuple is a 5-tuple.

The recurring structure observed in alarm information can therefore be represented as:

e = (System, Identity, DeviceType, Location, Condition)	**(1)**

The tuple provides a concise representation of relationships already present in the system being observed.

OBSERVATION BEFORE FORMALISM
The formal model provides a language that describes a pattern that I had already encountered repeatedly in working systems.

The five-role teaching model used in The Grammar of a Fire Alarm System corresponds to PANEL_ID, DEVICE_ID, DEVICE_TYPE, DEVICE_LOCATION, and the reported-condition role. SAFER STATUS is a reporting code, not a one-to-one encoding of persistent condition.

1.1 Making an Implicit Structure Explicit

SAFER does not introduce these facts into fire-alarm practice. The industry already uses them implicitly every day. A point list relies on identity, device type, and location. A device count groups points by device type. A floor plan relates a point to a place. An event history adds condition over time. A receiving system adds source context.

The useful step is to recognize that these are recurring semantic roles rather than columns that belong only to one manufacturer, database, drawing, or display. Once the roles are explicit, they can be compared, validated, serialized, and shared while each source system retains its native detail.

IMPLICIT → EXPLICIT
The five facts were already being used. SAFER names the recurring structure so the same meaning can cross system boundaries.

2. The Primitive Event Statement

The same structure can be expressed without mathematical notation. A reported event can be stated as:

PRIMITIVE EVENT STATEMENT *This system reports that this device, of this type, at this location, has changed to this state.*

Each phrase has a specific semantic role.

Role	SAFER field	Plain-language question
System	**PANEL_ID**	Who is reporting?
Identity	**DEVICE_ID**	Which point?
Device Type	**DEVICE_TYPE**	What kind?
Location	**DEVICE_LOCATION**	Where?
Reported condition	**STATUS (v1.1 reporting code)**	What is being reported?

The conceptual reported condition is broader than numeric SAFER STATUS. The optional CONDITION field can preserve more specific human-readable source detail without changing the reporting code. In current SAFER v1.1 serialization, STATUS combines several kinds of reporting meaning: condition/classification, restoration, repetition, or report purpose. A received message may describe a new transition, restore, repeat, or snapshot; it does not prove a new physical change occurred.

A particular reported event might therefore be represented as:

(Panel 01, L1S067, Smoke Detector, Classroom 4 West, ACTIVATED)	**(2)**

and expressed in ordinary language as:

Panel 01 reports that smoke detector L1S067 in Classroom 4 West has changed to ACTIVATED.

The mathematical tuple and the natural-language sentence describe the same relationships in different forms. This is one reason the term event grammar is useful. The individual values can change while their roles remain recognizable.

A waterflow switch can replace the smoke detector. A sprinkler room can replace the classroom. TROUBLE can replace ACTIVATED.

EVENT GRAMMAR
The values change. The grammatical roles remain.

3. The Stable Object and the Changing Condition

The five-part event representation contains an important internal distinction.

Three of the values describe the object itself:

$O = (Identity, DeviceType, Location)$	(3)

These properties connect the point to the building and its intended purpose. Identity tells us which point it is. Device Type tells us what the point represents. Location tells us where that point belongs in the building.

The point identifier is interpreted within its reporting system and relevant site context. Type and location are descriptive values; corrections or relocation do not automatically create a new point identity.

During ordinary operation, these properties are relatively stable. A smoke detector may remain installed for years while moving through many operational conditions without becoming a different object.

$$\text{NORMAL} \rightarrow \text{ACTIVATED} \rightarrow \text{NORMAL}$$

$$\text{NORMAL} \rightarrow \text{TROUBLE} \rightarrow \text{NORMAL}$$

The detector's condition has changed. Its identity, device type, and intended location have not.

BUILDING / CONFIGURATION CONTEXT $O = (I, T, L)$	→	OPERATIONAL STATE $C(t)$

The complete event representation can therefore be written:

$e_t = (S, I, T, L, C_t)$	(4)

or, equivalently:

$e_t = (S, O, C_t)$	(5)

where $O = (I, T, L)$.

This separation is useful because configuration change and operational change are different kinds of change. A change from NORMAL to ACTIVATED is an operational state change.

$$\text{NORMAL} \rightarrow \text{ACTIVATED}$$

Moving a detector to another room, replacing it with another device type, changing its programmed identity, or correcting its documented location is a configuration or documentation change. Those changes alter the description of the object. An activation reports what the object is doing.

PRECISE STATEMENT Identity, device type, and location describe the relatively stable configuration of a point within the building. They normally change when the building or system configuration changes, or when the recorded description is corrected or refined. Condition is the time-varying property observed during day-to-day operation.

This reflects an important relationship to building intent. The stable properties tell us what the thing is supposed to be and where it belongs. The condition tells us what that thing is reporting now.

3.1 The Condition Does Not Carry All of the Meaning

The same reported condition can mean different things when it belongs to different objects.

Consider two monitored dry contacts. Electrically, each may simply change from its normal state to an active state. One contact belongs to a manual pull station. The other belongs to a switch monitoring the position of a sprinkler control valve.

In both cases the underlying observation can be described as:

$$\text{CONDITION} = \text{ACTIVATED}$$

The meaning is different because the objects are different.

PULL STATION + ACTIVATED **Fire-alarm interpretation**	→	**VALVE SUPERVISORY SWITCH + ACTIVATED** **Supervisory interpretation**

The observed condition therefore should not be required to contain the meaning that already belongs to DEVICE_TYPE and system configuration.

OBJECT + CONDITION PROVIDE MEANING The same electrical change can have different significance because it happened to a different kind of object.

This is one reason the SAFER fields remain separate. DEVICE_TYPE answers what kind of thing is reporting. The underlying condition describes what changed. In the serialized SAFER representation, numeric STATUS can then carry the normalized event status used by receiving software.

4. State, Event, Classification, Integrity, and Response

A state and an event are related, but they are not identical.

A state describes what is true at a particular time. An event records or communicates a change or observation involving that state.

$$C(10{:}13) = \text{NORMAL}$$

$$C(10{:}14) = \text{ACTIVATED}$$

The transition can be written:

$$\text{NORMAL} \rightarrow \text{ACTIVATED}$$

The transition is the event of interest. ACTIVATED is the resulting condition.

This distinction becomes particularly important in fire alarm systems because an observed point condition does not by itself determine how the event will be classified or what response will follow.

Consider two objects:

$$O_1 = (\text{L1S067, Smoke Detector, Classroom 4 West})$$

$$O_2 = (\text{L1S082, Valve Supervisory Switch, Riser Room})$$

Both can experience the same point transition:

$$\text{NORMAL} \rightarrow \text{ACTIVATED}$$

Yet their resulting classifications can be different.

Object type	Condition	Resulting classification
Smoke Detector	ACTIVATED	Fire Alarm
Valve Supervisory Switch	ACTIVATED	Supervisory
Monitored system or pathway	TROUBLE	Trouble

The relationship can be expressed conceptually as:

$$\text{Object} + \text{Condition} \rightarrow \text{Classification}$$

or more formally:

$$K_t = f(e_t, \text{system configuration, receiving context}) \quad (6)$$

Classification is therefore treated as a derived layer rather than one of the five primitive event facts. The applicable system configuration and receiving context can affect whether an observed condition is handled as alarm, supervisory, trouble, or another category.

OBSERVATION ≠ CLASSIFICATION ≠ RESPONSE ACTIVATED describes the reported point condition. A classification such as FIRE ALARM or SUPERVISORY describes how that observation is interpreted within the system. The response is a separate application decision made according to configuration, plan, policy, and authority.

A receiving application can determine a response without changing the underlying observation:

$$R_t = g(e_t, K_t, \text{plan, operational policy}) \quad (7)$$

4.1 One Circuit Can Carry Different Meanings

A common fire-alarm example makes the distinction visible.

An addressable monitoring module on a signaling line circuit (SLC) can supervise an initiating device circuit (IDC) containing a dry contact and an end-of-line (EOL) resistor.

Under normal conditions, the module sees the expected circuit condition. When the dry contact operates, the electrical condition changes in the intended way. If the circuit becomes open or otherwise loses the expected supervised electrical condition, the module can report trouble.

The wiring arrangement does not by itself determine what the contact means.

Monitored item	Observed change	Application meaning
Manual pull station	Dry contact operates	Fire alarm
Sprinkler control-valve switch	Dry contact operates	Supervisory
IDC / monitoring path	Expected supervision is lost	Trouble

The same basic monitoring mechanism therefore supports three conceptually different statements: the monitored object operated; the meaning assigned to that operation depends on what the object is; and the monitoring path itself might become unreliable.

IMPORTANT DISTINCTION In this monitoring example, a supervisory signal says something about the condition of the thing being watched. A trouble signal says something about our ability to watch it.

A closed sprinkler control valve can be reported correctly by a perfectly functioning monitoring circuit. The resulting supervisory signal is meaningful information from the monitored object.

An open or failed monitoring circuit presents a different problem. The system might no longer be able to determine the condition of the valve with confidence. That is an integrity problem.

4.2 A Practical Implementation Pattern: Base State and Logical State

RescueLogic provides one practical example of this separation. It distinguishes the basic operational state recognized from an incoming condition from the logical category used by the HMI. These are RescueLogic application layers, not additional SAFER fields.

Layer	Values	Purpose
Base State	NORMAL • ALARM • TROUBLE	Reduces manufacturer-specific conditions to three fundamental operational states.
Logical State	1 NORMAL • 2 FIRE ALARM • 3 TROUBLE • 4 SUPERVISORY ALARM	Uses device meaning and site configuration to categorize the point for the application and HMI.

This allows the same underlying active condition to be presented differently according to what the monitored object represents.

Example	Base State	Logical State
Pull station operates	ALARM	FIRE ALARM
Valve supervisory switch operates	ALARM	SUPERVISORY ALARM
Supervised circuit fault	TROUBLE	TROUBLE

The distinction can inform SAFER translation without making the numeric vocabularies interchangeable. A pull-station activation may be reported using SAFER STATUS = 1 (Active), while an activated valve-supervisory

point may use STATUS = 2 (Supervisory). The reporting code expresses SAFER meaning; it is not a numerical copy of the RescueLogic Base State or Logical State key. Point identity, type, and location remain separate.

The HMI therefore does not have to infer meaning from the electrical condition alone. It combines the recognized condition with the type and configuration of the object.

IMPLEMENTATION LESSON Base State ALARM is an internal active-state category, not necessarily the final Fire Alarm classification. Logical State applies device meaning and site configuration to determine how the HMI categorizes the point.

Event Type, Device Type, and Descriptive Condition

RescueLogic's Script Panel Types configuration is a practical example of separating broad state recognition from detailed source information. The Event Type tab recognizes incoming messages associated with the three Base States: Normal, Alarm, and Trouble. The Device tab extracts device type, address, and descriptive Condition text from the source message.

Device State Images provide a recognizable visual representation associated with the device type and its state. Descriptive Condition text can explain the reported situation more specifically, including a symptom or clue about a trouble condition.

Multiple messages can therefore share the same broad Trouble state while retaining different descriptions, such as Open Circuit, Ground Fault, or Detector Missing. The three Base States remain simple and recognizable while descriptive conditions remain open-ended.

This is a RescueLogic implementation pattern, not a required SAFER presentation. SAFER's six STATUS reporting codes are not numerically equivalent to RescueLogic's three Base State keys or four Logical State categories.

4.3 Two Different Questions

A supervised system is continually answering more than one kind of question.

OBJECT QUESTION **What is this object reporting?**	→	**INTEGRITY QUESTION** **Can the system still be relied upon to tell us?**

Those questions should remain separate. A valve supervisory switch can be ACTIVATED while the circuit monitoring it remains completely healthy. Conversely, the valve might not have changed position at all while a wiring or communications problem causes the system to lose confidence in its ability to observe the valve.

This produces two independent kinds of information:

OBJECT CONDITION **What is the thing reporting?**	→	**MONITORING INTEGRITY** **Can we still rely on the observation?**

The distinction is important because the last known state of an object is not necessarily its present state after supervision has been lost.

DO NOT MANUFACTURE CERTAINTY A system should not silently turn uncertainty into certainty.

4.4 YES, NO, and MAYBE

Many practical questions do not require another detailed classification. They require a simple answer.

Examples include: Will it work? Does this information agree? Is this point where the documentation says it is? Do we have enough information to rely on this conclusion?

Answer	Meaning
YES	The available information supports the proposition.
NO	The available information contradicts the proposition.
MAYBE	The available information does not justify either YES or NO.

A derived three-state answer can be represented as:

$$A(q, e_t, \text{context}) \in \{\text{YES, NO, MAYBE}\} \quad (8)$$

where q is the question being asked and context represents any additional information needed to answer it.

THREE-STATE PRINCIPLE Maybe makes Yes and No more trustworthy.

MAYBE is not a weaker form of YES or NO. It protects the accuracy of the other two answers. If incomplete, conflicting, or unreliable information were forced into a binary answer, YES and NO would necessarily become less trustworthy.

YES, NO, and MAYBE are answers derived by a receiving application for a particular question. They are not additional required SAFER v1.1 fields or replacement STATUS codes.

Fire-alarm supervision supplies useful but bounded evidence. A normal monitored path can support YES to the narrower question 'Is this supervision indication normal?', but cannot alone prove YES to 'Will the complete protected function work when needed?' Evidence of an unavailable required function may support NO; otherwise TROUBLE often warrants MAYBE until testing or additional evidence resolves uncertainty. The reported TROUBLE remains source information; MAYBE is the derived answer to the particular question.

The same evidence can support different answers to different questions. Suppose two records describe the same DEVICE_ID. One calls it a Smoke Detector and the other calls it a Pull Station.

Question	Answer
Do the two records agree about DEVICE_TYPE?	**NO**
Is the Smoke Detector record proven correct by the available evidence?	**MAYBE**
Is the Pull Station record proven correct by the available evidence?	**MAYBE**

The available evidence proves disagreement, but it does not yet prove which record is correct. No additional uncertainty has been created. The three-state answer simply prevents the application from claiming more than the available evidence supports.

4.5 Why the Separation Matters in Practice

In day-to-day fire alarm work, the same reported event can be viewed differently depending on who is using the information.

A technician may need to know exactly what the point reported and whether its communications path is healthy. An operator may need to know how the event has been classified. A responder may need to know what action the site's procedures require. A validation program may need only to answer whether two sources agree. An automated or AI-assisted system may need to distinguish what is known from what is inferred.

These are related questions, but they are not the same fact.

KEEP THE SOURCE FACTS VISIBLE Preserve the source event. Add interpretation without replacing the original observation.

This separation allows interpretations to change as additional information becomes available while keeping the source record stable and reviewable.

4.6 From Information to Human Recognition

At the site where decisions are made, a recognizable device icon or image, its reported condition, and a known location help people form a mental picture. The device type connects the present report with experience and expected behavior. This presentation supports recognition and communication without replacing the source observation, classification, or applicable response procedures.

4.7 Distinguishing Report Identity from Event Identity

A smoke detector may first report activation at 10:00 and again report that the condition remains present at 10:05. The two reports can refer to the same point and continuing condition without being the same transmitted message. Message identity, source event correlation, and physical transition identity are distinct concepts. SAFER_ID.unique_id should not automatically be assumed to identify an entire continuing condition. Future design review should test duplicate, repeated, out-of-order, and restored reports without prematurely mandating a new wire field.

5. Information Model Before Message Format

Once the event has been expressed formally, another distinction becomes possible.

The information model and the communications format are separate things. The model is:

$e_t = (\text{System}, \text{Identity}, \text{DeviceType}, \text{Location}, \text{Condition}_t)$	(9)

SAFER can serialize parts of the model into JSON. Its STATUS field is a normalized reporting code, rather than a universally equivalent encoding of persistent condition. This excerpt illustrates the five information roles and deliberately omits other required baseline fields:

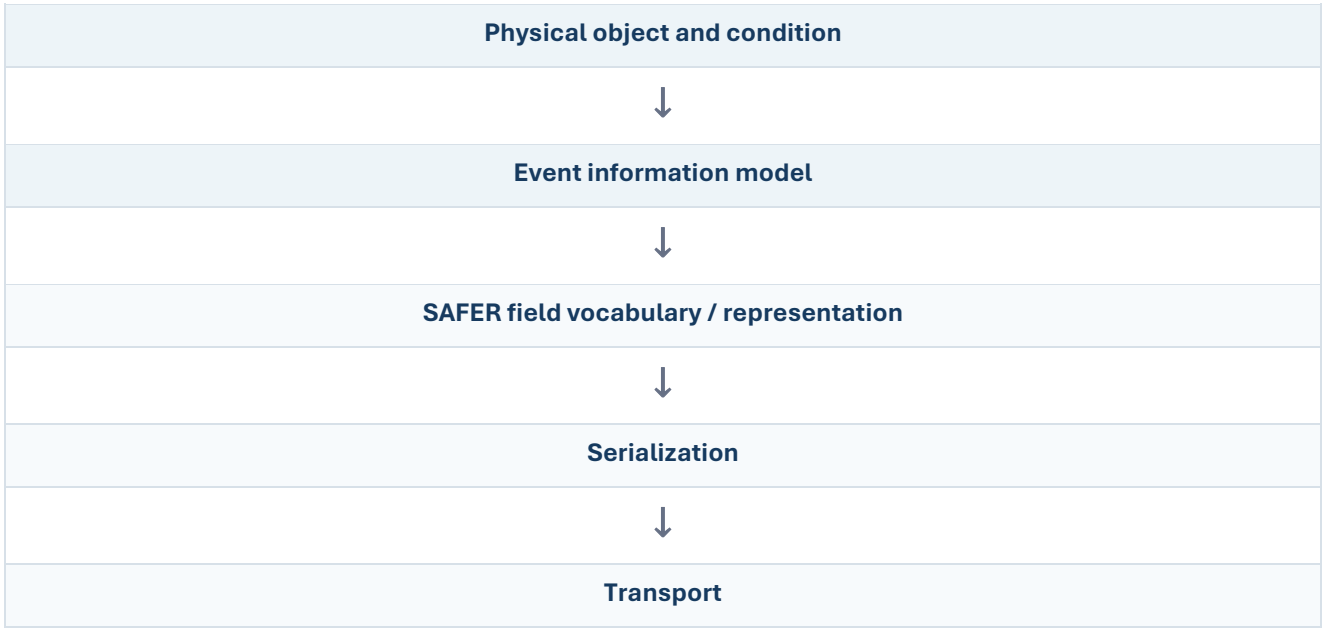
```
{
  "PANEL_ID": "PANEL 01",
  "DEVICE_ID": "L1S067",
  "DEVICE_TYPE": "SMOKE DETECTOR",
  "DEVICE_LOCATION": "CLASSROOM 4 WEST",
  "STATUS": 1
}
```

MODEL BEFORE SERIALIZATION

The information model defines these relationships. JSON carries them.

This produces a useful layered view:

The subscript t represents event time in the abstract model. A SAFER implementation can also carry an explicit timestamp as message metadata for history, auditing, and correlation. That timestamp records when an event was observed or reported; it does not replace any of the five core semantic roles.



A different serialization or transport could theoretically carry the same information model. This is significant for interoperability because it separates the meaning of the event from the technology used to transmit it.

Information Model ≠ JSON Serialization

Two systems can share an understanding of System, Identity, DeviceType, Location, and Condition even when their internal architectures are different.

5.1 Numeric STATUS as a Stable Machine Value

In the abstract model, Condition is the changing semantic role. In the serialized SAFER vocabulary, STATUS is a defined numeric event-status code. The number is intended to remain stable across applications, databases, languages, and transports even when the human-facing wording changes.

The original SAFER design began with the familiar compact numeric pattern used in traditional alarm reporting and extended it for supervisory, trouble, and status handling. SAFER v1.1 defines:

Code	SAFER meaning
1	Active
2	Supervisory
3	Restore
4	Trouble
5	Status only
6	Previously reported

Some values describe an observed non-normal state; others describe how an event is being reported. RESTORE reports restoration toward normal; receipt alone does not independently verify the physical transition, while STATUS ONLY and PREVIOUSLY REPORTED distinguish information that should not necessarily be treated as a newly occurring event. For that reason, STATUS is best understood as the normalized event-status code in the SAFER representation, while Condition remains the abstract semantic role in the five-part model.

STATUS = 5 (Status only) and STATUS = 6 (Previously reported) usually provide information rather than announce a new activation requiring immediate action. At sites that expect recurring reports, a receiving application can compare arrivals with a documented schedule. A missing expected report may warrant investigation of reporting continuity; absence alone does not establish the point's physical condition. These expectations and actions are separate from the SAFER STATUS code.

Numeric Reporting Meaning and Descriptive Detail

SAFER's required numeric STATUS communicates a defined reporting meaning. The optional CONDITION field can preserve more specific human-readable information about what the source reported.

For example, Open Circuit and Ground Fault may both accompany SAFER STATUS = 4 (Trouble). A receiving application may present both through a recognizable Trouble state while preserving the different descriptions for operators, technicians, and diagnostic work.

CONDITION does not override STATUS, and descriptive source information does not necessarily establish the physical cause of a reported trouble. These separate information roles allow a compact reporting vocabulary without restricting the variety of conditions future devices may describe.

STABLE MACHINE VALUE A query such as STATUS = 1 has one defined meaning. An HMI can display ACTIVE, a localized label, or another presentation without changing the transmitted value.

A stable numeric code supports reproducible decoding within a defined protocol version. It does not by itself preserve every dimension of a source report: classification and a qualifier such as 'previously reported' may

both apply. Adapters should retain original source context when possible, and future examples and tests can help evaluate whether a later version should separate these dimensions.

These protocol labels do not redefine established fire-alarm signal categories. For example, STATUS = 2 is the SAFER machine value for a supervisory event status; a receiving application can still preserve source detail and apply its own classification and response logic.

5.2 A Deliberate Field-Naming Convention

SAFER uses uppercase field names deliberately:

```
PANEL_ID
DEVICE_ID
DEVICE_TYPE
DEVICE_LOCATION
STATUS
```

The convention serves both human and machine readers. In the fire alarm and building-documentation environment, uppercase identifiers are visually familiar. Alarm-panel displays, equipment labels, point identifiers, and traditional engineering drawings commonly use uppercase text to distinguish technical information from ordinary prose. SAFER carries that visual language into structured data.

The convention also gives defined life-safety information a recognizable appearance when SAFER fields are placed beside ordinary application data:

Illustrative application wrapper: the nested SAFER object carries the baseline payload; the outer applicationName and lastUpdate belong to the host application.

```
{
 "applicationName": "Building Monitor",
 "lastUpdate": "2026-09-17T09:14:22-04:00",
 "SAFER": {
  "CUSTOMER": "DemoSite",
  "PANEL_ID": "PANEL 01",
  "DEVICE_ID": "L1S067",
  "DEVICE_TYPE": "SMOKE DETECTOR",
  "DEVICE_LOCATION": "CLASSROOM 4 WEST",
  "STATUS": 1,
  "SAFER_ID": {
   "version": "1.1",
   "unique_id": "123e4567-e89b-42d3-a456-426614174001"
  }
 }
}
```

To a person reading the message, the SAFER vocabulary is immediately visible. To software, the same convention provides a convenient syntactic cue. A program can test directly for known field names such as PANEL_ID, DEVICE_ID, and STATUS, separating the SAFER vocabulary from unrelated application properties.

Semantic identity comes from the defined SAFER field names and schema; uppercase provides an additional visual and syntactic cue. Software should recognize the defined SAFER field names and schema, rather than infer that any uppercase value belongs to SAFER.

NAMING LAYER

The information model defines what a field means. The SAFER vocabulary defines what that role is called. The uppercase naming convention helps people and software recognize it when they see it.

1	2	3
Continuity with the alarm industry	**Continuity with drawings**	**Visual separation**
Panel messages, point identifiers, annunciator text, and technical labels often appear in uppercase.	Traditional architectural and engineering drawings have a long visual tradition of uppercase notation.	DEVICE_LOCATION reads as a defined life-safety data element rather than an ordinary application variable such as deviceLocation.

Uppercase is a representation convention, not part of the abstract five-role model.

6. Requirements for Interoperability

A fire alarm system is not a single information environment.

The same physical point may exist simultaneously in several forms:

- as a programmed point in a fire alarm control unit;
- as a row in a database;
- as a symbol on a floor plan;
- as an active event in an operator display;
- as a line in an event history;
- as a transmitted message; and
- as information ultimately presented to a responder.

Each representation may have a different purpose. A technician may need an exact point address. An operator may need a useful description. A responder may need a destination and orientation within the building. A receiving computer needs structured information with defined meaning that can be interpreted consistently.

SYNTACTIC QUESTION **Can one computer send and parse the data?**	→	**SEMANTIC QUESTION** **Will the receiving system understand what the data means?**

Two systems can successfully exchange bytes while still disagreeing about the meaning of those bytes. This distinction is often described as the difference between syntactic interoperability and semantic interoperability.

Syntactic interoperability concerns whether systems can exchange and parse information in a compatible format. Semantic interoperability concerns whether the systems share enough understanding of the information for the received data to retain its intended meaning.

That distinction is well established in existing IoT work. For example, oneM2M distinguishes syntactic from semantic interoperability and provides a minimal Base Ontology intended to help outside information models map into a common semantic framework. [2, 3]

SAFER approaches the same general problem from a narrower life-safety perspective. It asks:

INTEROPERABILITY QUESTION *What is the smallest useful structure that communicates the basic meaning of a reported event?*

$$e_t = (\text{System}, \text{Identity}, \text{DeviceType}, \text{Location}, \text{Condition}_t)$$

COMMON BOUNDARY

The objective is to give them a common semantic point of contact rather than forcing each participating system to conform internally.

6.1 Keeping SAFER Terms and NFPA Terms in Their Proper Roles

Fire alarm professionals already work with several different kinds of information that can sound similar when placed in one data message. A device type, a point condition, a signal classification, and a required response each answer a different question.

SAFER keeps those roles separate so information can be exchanged without changing the meaning of established fire alarm terms.

Question	Information
What kind of thing is it?	DEVICE_TYPE — Smoke Detector, Waterflow Switch, Valve Supervisory Switch
What is it reporting?	STATUS — numeric event-status code (for example, 1 Active, 2 Supervisory, 4 Trouble)
How is the event being handled?	Classification — Fire Alarm, Supervisory, Trouble
What should be done?	Response — site-specific procedures, operator or responder action

The same word can appear at more than one layer without having the same role. SUPERVISORY, for example, may be the human label associated with STATUS = 2 in a SAFER message and also a signal classification used by the receiving application. The numeric STATUS is the machine value carried with the event; classification and response remain application decisions.

NFPA 72 uses related concepts in different contexts, but it does not define the SAFER five-field model. The purpose of mentioning NFPA 72 here is to remind the reader that established fire-alarm terminology already distinguishes concepts such as source, initiating device, location, signal classification, changes of state, and response requirements. SAFER is designed to preserve those distinctions with clarity. [8]

SAFER PRESERVES ESTABLISHED FIRE-ALARM TERMINOLOGY

NFPA 72 and SAFER serve different purposes. SAFER does not redefine alarm, supervisory, trouble, initiating device, location, or response requirements. It provides a common structure for carrying information while allowing those established meanings and project requirements to remain authoritative.

Requirements for a particular installation should be verified against the applicable edition of NFPA 72, manufacturer documentation, and the requirements of the Authority Having Jurisdiction.

6.2 When the Source Knows More Than the Message Carries

Modern intelligent fire-alarm systems often know more about a point than a compact supervising-station report carries. That is not a defect in the older format. Contact ID solved an important interoperability problem with a compact DTMF reporting structure appropriate to its design environment, and the format remains widely supported. [10]

That distinction begins with identification. In the Ademco No. 678, the programming form treated the primary telephone number, secondary telephone number, and subscriber identification number as separate entries. Its operating description says the communicator dialed a preprogrammed telephone number and, after receiver handshake, transmitted a message containing the subscriber identification number and alarm code. [9] The subscriber number therefore identified the account in the transmitted report; it was not part of the telephone dialing mechanism or a callback number.

Contact ID uses a compact reporting structure built around subscriber/account identification, an event qualifier and event code, group or partition context, and a contact/zone/user value. Honeywell Fire-Lite documentation shows this as a subscriber ID account code followed by Contact ID identification, event qualifier, event code, group number, and zone/point number. [13] Contact ID Q=1 indicates a new event/opening; Q=3 a restore/closing; Q=6 a previously reported condition still present. Q is distinct from event code and partition-zone/user identification; none of these qualifier numbers is automatically a SAFER STATUS mapping.

Honeywell Silent Knight documentation makes the difference especially clear by showing SIA and Contact ID reporting side by side. Its SIA fixed-length structure can carry panel ID, SBUS ID, zone, point, group, emergency, and receiver information, while its Contact ID columns are qualifier, event code, group, and contact. [12]

A modern addressable system can therefore have a native hierarchy such as panel or node N01 with point L1S001 or L1M001, while a Contact ID message has no dedicated semantic Panel ID field corresponding to that hierarchy. A receiver or integration may use account context, the contact/zone value, or external mapping to recover additional meaning. The 2016 SIA committee record notes that a 10-digit Contact ID option was added to DC-05, increasing identifier capacity without by itself defining a hierarchy such as Site → Panel → Point. [10, 11]

MORE DIGITS ≠ MORE SEMANTIC STRUCTURE

Increasing the capacity of an identifier creates more possible values. It does not define what those values mean.

Modern intelligent systems can retain richer source information. Kidde/Fenwal SmartOne documentation describes a programmable system address, a reporting type, and a human-readable owner location message. [14] Those facts can be mapped into SAFER roles without requiring the source system to abandon its native representation.

The following comparison is illustrative. Individual manufacturer, communicator, receiver, and automation implementations can vary.

Information known at the source	Legacy reporting boundary	SAFER
Panel / node N01	No dedicated Contact ID Panel ID field	PANEL_ID = N01
Native point L1S001	Contact/zone value or external mapping	DEVICE_ID = L1S001
Smoke Detector	Event-code/receiver interpretation or lookup	DEVICE_TYPE = SMOKE DETECTOR
Classroom 4 West	No dedicated Contact ID human-location field	DEVICE_LOCATION = CLASSROOM 4 WEST
ACTIVATED	Qualifier/event-code semantics	STATUS = 1 (Active)

A mapping database is not inherently a bad design. The difficulty arises when meaning is distributed across separate records that must remain synchronized. A point number can function as an opaque key: it identifies a record, but its human meaning has to be recovered somewhere else.

The source panel may contain device details that the transmitted Contact ID message does not carry. Other details may come from documented configuration or maintained customer records. A translation should distinguish transmitted information from information obtained through lookup or interpretation.

MODERN TRANSPORT DOES NOT AUTOMATICALLY PRODUCE A MODERN INFORMATION MODEL

IP or cellular conversion can modernize the communications pathway while retaining the older event representation. NFPA committee-development material describes IP dialer-capture modules that take a DACT output and convert its data stream to IP, and cellular devices that accept DACT tone signaling and packetize it for a wireless network. [15]

This is a useful migration technique: the transport can change while the event representation remains the same. SAFER addresses a different layer — the shared meaning carried between systems.

NFPA 72 addresses changes to site-specific software and documentation, and committee-development material specifically discusses testing and record updates for software changes, including customized descriptive location-label changes. [16] It does not define the SAFER-style shared information model by which independent representations of point identity, device type, location, and condition remain synchronized across systems.

THE SOURCE MAY KNOW MORE THAN THE REPORT CARRIES

The translation identifies the source of each fact and makes the resulting information understandable.

7. A Common Layer Across Different Internal Architectures

Different manufacturers organize their systems in very different ways. One may use loops and devices. Another may use nodes, cards, circuits, modules, sensors, or logical points. The detailed internal structures are important for installation, testing, service, diagnostics, and manufacturer-specific capabilities.

A system can retain its native information while also presenting the event through a small shared structure. Manufacturer-specific information can then accompany that common representation when additional detail is useful.

System	Identity	Device Type	Location	Condition

ARCHITECTURAL PRINCIPLE

Interoperability requires agreement at the boundary where information is exchanged while allowing internal architectures to remain different.

Manufacturer A	→ SAFER Event Meaning →	Applications and People
Manufacturer B		
Manufacturer C		

The source system remains authoritative for its native information. SAFER provides a common description of the reported event.

The common layer does not need to contain every fact known by every manufacturer. Its role is to preserve the basic meaning that must survive the handoff.

8. Related Information Models in IoT and Industrial Systems

Separating information meaning from transport technology is an established principle in Internet of Things and industrial-system architecture. SAFER applies that principle to the life-safety event domain. These comparisons locate SAFER architecturally and show how a domain-specific life-safety model can complement broader interoperability frameworks.

These technologies provide broader interoperability mechanisms, but they do not themselves define the small life-safety event vocabulary proposed here.

Technology	SAFER Integration
W3C Web of Things	SAFER can serve as the life-safety event payload structure for a fire-alarm panel or logical point described by a WoT Thing Description. WoT describes the interface and protocol binding; SAFER supplies the event meaning. [1]
oneM2M	SAFER can map into oneM2M through its ontology and interworking mechanisms, while serialized SAFER data travels through the oneM2M service layer and its HTTP/MQTT/CoAP bindings. [2, 3]
OPC UA	SAFER can define a small domain-specific information model or EventType within an OPC UA system. OPC UA provides the broader object model, services, subscriptions, security, and transport; SAFER supplies the life-safety event meaning. [4]
OGC SensorThings	SAFER can be associated with a Thing or sensing point and exposed as an Observation or structured result, particularly where location and geospatial relationships matter. SensorThings provides the Web/API and geospatial framework; SAFER preserves the life-safety event semantics. [5]
MQTT	MQTT carries a serialized SAFER message. SAFER supplies meaning; MQTT supplies message transport. [6]
Sparkplug	SAFER can map into Sparkplug metrics or travel as application-specific structured content within a Sparkplug/MQTT environment. Sparkplug provides topic structure, payload conventions, and session-state behavior; SAFER supplies the life-safety event meaning. [7]

8.1 W3C Web of Things

The W3C Web of Things defines a Thing Description as a formal information model and common representation for describing physical or virtual Things and the ways software can interact with them. A Thing Description can describe metadata, properties, actions, events, data schemas, security information, and protocol bindings. Its standard representation uses JSON with JSON-LD processing support, while the specification distinguishes the information model from its representation. [1]

Information Model ≠ JSON Serialization

W3C WoT ASKS **How can software understand and interact with a Thing?**	→	**SAFER ASKS** **What does this reported life-safety event mean?**

A SAFER representation could therefore complement a broader Thing Description rather than substitute for it.

8.2 oneM2M

oneM2M addresses interoperability across heterogeneous IoT systems. Its Base Ontology is intentionally minimal so external ontologies can be mapped into a common semantic framework, while its ontology-based

interworking specification describes how external systems can interwork with oneM2M through ontology mappings. [2, 3]

RELEVANT DESIGN PARALLEL

A common model can remain relatively small while allowing richer outside models to retain their own detail.

SAFER applies a comparable principle within a narrower domain. Manufacturer-specific information can remain intact while the five event roles provide a small shared structure for mapping system-specific meanings.

8.3 OPC UA

OPC Unified Architecture is a platform-independent industrial information-exchange architecture. It explicitly distinguishes an information model, a message model, a communication model, and a conformance model. The information model represents structure, behavior, and semantics; other layers address application interaction, transfer, and interoperability. [4]

Event Meaning → SAFER Information Model → Message Representation → Transport

OPC UA is much broader than SAFER and supports extensive industrial modeling and control. The relevance here is architectural rather than competitive: information meaning can be defined separately from the mechanism that carries it.

8.4 OGC SensorThings

The OGC SensorThings API provides a standardized, Web-based way to connect heterogeneous IoT devices, observations, and applications. Part 1: Sensing defines common entities for Things, Locations, Datastreams, Sensors, ObservedProperties, Observations, and FeaturesOfInterest; Part 2 addresses tasking. [5]

SensorThings is relevant because its observation model makes relationships explicit instead of collapsing them into an unstructured message. SAFER serves a different domain and vocabulary, but the interoperability lesson is similar.

SHARED INTEROPERABILITY LESSON

Information becomes more portable when the roles within the information are explicitly named.

8.5 MQTT and Sparkplug

MQTT is a lightweight client/server publish-subscribe messaging transport and does not, by itself, define the application semantics of the payload it carries. Eclipse Sparkplug adds an MQTT topic namespace, payload conventions, and session-state management for industrial and real-time SCADA environments. [6, 7]

Transport ≠ Meaning

A transport can deliver a message successfully while leaving its application meaning undefined. SAFER concentrates on the semantic content of the life-safety event and leaves transport to established communications mechanisms.

8.6 Integration Patterns: Contained, Mapped, or Transported

The technologies above operate at different architectural layers, so SAFER would not integrate with each in the same way. Its five-role event model can be contained within a broader information model, mapped into another semantic framework, or serialized and transported by an established messaging system.

Pattern	What SAFER does	What the surrounding architecture does
Contained	SAFER becomes a domain-specific event structure inside a larger information model.	Provides the larger Thing/object model, interfaces, security, services, and communications behavior.
Mapped	SAFER roles are related to equivalent concepts in another semantic model.	Provides cross-domain semantics and ontology/interworking mechanisms.
Transported	A serialized SAFER event is carried as application data.	Provides messaging, routing, delivery, session state, or transport.

The likely relationship depends on the role each surrounding technology is already designed to play.

Framework	Likely relationship	Practical interpretation
W3C WoT	Contained / transported	A Thing Description can describe a Thing and its Events; SAFER can supply the life-safety event content while WoT describes interfaces, bindings, and security.
oneM2M	Mapped / transported	SAFER concepts can map into the broader semantic framework while oneM2M provides service-layer and protocol-binding mechanisms.
OPC UA	Contained / mapped	SAFER can inform a life-safety EventType or companion model while OPC UA provides the larger industrial information model and services.
OGC SensorThings	Mapped / contained	SAFER can be associated with a known Thing, location, or observation where sensing and geospatial context are useful.
MQTT	Transported	MQTT carries the message; SAFER supplies its life-safety meaning.
Sparkplug	Mapped / transported	Sparkplug supplies MQTT topic, payload, and session conventions; SAFER contributes the life-safety event semantics.

These relationships follow the architectural roles described above: WoT defines Things, Events, and bindings [1]; oneM2M provides ontology mapping and interworking mechanisms [2, 3]; OPC UA exposes EventTypes within a broader information model [4]; SensorThings organizes Things, Locations, Datastreams, and Observations [5]; and Sparkplug adds topic, payload, and session conventions to MQTT [7].

Manufacturer / Source System
↓
SAFER Event Meaning **System • Identity • DeviceType • Location • Condition**
↓
Mapping / Representation **WoT • oneM2M • OPC UA • SensorThings • Sparkplug**
↓
Transport / Applications **MQTT • HTTP • other networks • software • people**

SAFER provides the domain meaning. Other technologies may contain it, map it, or carry it.

9. Where SAFER Fits

These existing approaches operate at different levels and solve different problems.

PROPOSED ROLE *SAFER is a domain-specific event information model for communicating the basic meaning of a reported life-safety event.*

SAFER remains focused on the meaning of the reported event. Surrounding IoT and communications architectures can provide device capabilities, networking, security, discovery, databases, and presentation.

FOCUS QUESTION *When a life-safety system reports an event, what information should cross the boundary with it?*

$$e_t = (\text{System}, \text{Identity}, \text{DeviceType}, \text{Location}, \text{Condition}_t)$$

This small structure sits between the source system and the systems or people that need to understand the event.

Physical Building
↓
Manufacturer / Source System
↓
SAFER Event Model
↓
Operator • Technician • Floor Plan • Responder • Computer

The five core roles provide a shared center. Different consumers can then present different views of the same underlying facts. That relationship is already visible in RescueLogic, where a single point can appear in a list, on a floor plan, in event history, in a report, in site-specific response information, or in a technical view while the underlying device information remains consistent.

10. Carrying Context With the Event

Historically, communications channels often imposed severe limits on message size. Compact signaling methods therefore had an important practical advantage: a short code could efficiently report an event over a constrained channel.

The tradeoff is that some of the human meaning may reside outside the transmitted message. A receiving system may need an external record to determine:

- Which point does code 067 represent?
- What type of point is it?
- Where is it?

If the transmitted event contains only an identifier, the receiving application must recover the remaining context elsewhere.

SAFER takes advantage of modern data transport by allowing more of that context to travel with the event:

```
{
  "CUSTOMER": "DemoSite",
  "PANEL_ID": "PANEL 01",
  "DEVICE_ID": "L1S067",
  "DEVICE_TYPE": "SMOKE DETECTOR",
  "DEVICE_LOCATION": "CLASSROOM 4 WEST",
  "STATUS": 1,
  "SAFER_ID": {
    "version": "1.1",
    "unique_id": "123e4567-e89b-42d3-a456-426614174000"
  }
}
```

COMPACT REFERENCE Reference to meaning	→	STRUCTURED EVENT More of the meaning carried with the message

Here, STATUS = 1 has the defined SAFER meaning Active. The receiving interface can display a human-readable label without requiring the source to transmit that wording.

Shared schema knowledge remains necessary. A receiving system must still understand the SAFER schema, field definitions, versions, and any coded enumerations. But carrying explicit identity, device type, location, and condition reduces the amount of event meaning that must be reconstructed from separate databases.

That has practical consequences for translation, auditing, troubleshooting, system integration, historical review, and presentation to people.

It also makes disagreements easier to discover. If two sources both refer to L1S067 but one describes it as a smoke detector and the other as a pull station, the software does not need to know immediately which description is correct.

Smoke Detector ≠ Pull Station

A USEFUL RESULT

The disagreement itself has become visible.

11. The Object, the Building, and Time

A life-safety point can be viewed as having two different classes of information.

BUILDING / CONFIGURATION CONTEXT (Identity, DeviceType, Location)	→	OPERATIONAL STATE Condition(t)

The first group tells us how the point belongs in the building. The second tells us what is happening to that point at a particular time. This distinction reflects the practical behavior of building systems.

A smoke detector installed in Classroom 4 West may remain associated with the same identity, device type, and location through thousands of ordinary observations. Those properties normally change only because something about the building, system, or description changes:

- the point is relocated;
- the device is replaced or reprogrammed;
- the system is renovated;
- the documented type is corrected; or
- the location description is refined to represent the building more accurately.

Condition behaves differently. It exists explicitly as a function of time:

$$C(t)$$

The point can therefore remain the same recognizable object while its operational condition changes:

$$\text{NORMAL} \rightarrow \text{ACTIVATED} \rightarrow \text{NORMAL}$$

The event model captures that changing relationship without requiring the object's stable description to be reconstructed for every interpretation. This gives SAFER a connection to both building intent and event time.

STABLE INFORMATION ANSWERS What is this thing, and where does it belong?	→	CHANGING INFORMATION ANSWERS What is it reporting now?

The emerging architecture

OBJECT	(Identity, DeviceType, Location)
↓	
CONDITION	C(t)
↓	
EVENT	(System, Object, $Condition_t$)
↓	
SAFER	Machine-readable representation of event meaning, including numeric STATUS

DERIVED APPLICATION LAYERS
Classification: $K_t = f(e_t$, system configuration, receiving context)
Response: $R_t = g(e_t, K_t$, plan, operational policy)
Three-state answer: $A(q, e_t$, context) ∈ {YES, NO, MAYBE}
These interpretations can change with context or additional information without rewriting the five core facts reported by the source.

12. Next Development of the Model

GitHub discussion can compare retaining the six combined SAFER STATUS codes with separating reported condition/classification from an event qualifier. Simplicity and backward compatibility, as well as precision and scalability, deserve consideration. Examples, source messages, expected interpretations, and conformance tests can be shared through GitHub Issues; specific changes can be proposed through Pull Requests for future versions.

Review question: How should broad state classifications and descriptive CONDITION information work together? A compact state vocabulary can remain recognizable while preserving specific condition details useful to different applications and people. Reviewers should test actual source messages and describe information preserved or lost before proposing a format change.

This Revision 0.2 working paper explains the event grammar, the stable-object / changing-condition distinction, the separation between information model and serialization, the numeric STATUS representation, the distinction between observation, classification, integrity, and response, the RescueLogic Base State / Logical State implementation pattern, and the derived YES / NO / MAYBE answer layer.

The next development of the paper can test and extend the model through:

1. Worked examples showing the same five-role structure across smoke detection, waterflow, supervisory, trouble, special-hazards, and non-fire building-system events.
2. Human and machine views showing how technicians, operators, responders, floor plans, histories, and receiving computers can present different views of the same underlying event facts.
3. Design principles for preserving original manufacturer information while mapping native events into the common semantic layer and numeric STATUS vocabulary.
4. A standards crosswalk that asks focused questions of the applicable NFPA 72 edition without reproducing copyrighted code text.
5. Validation cases for YES / NO / MAYBE answers, including incomplete, conflicting, stale, and newly restored information.
6. Versioning and extension rules for STATUS codes, preservation of source messages, auditability of derived interpretations, and mapping SAFER into broader IoT information models.

WORKING CONCLUSION SAFER makes explicit a structure the fire-alarm industry already uses implicitly every day. Different systems do not have to work the same way to share the same basic information. Meaning can remain stable while transport, serialization, presentation, and application logic evolve.

Related publications and discussion

SAFER Protocol v1.1, the Common Vocabulary, related publications, and opportunities for implementation feedback are available at:

https://github.com/rescuelogic/SAFER

This technical working paper offers discussion and explanatory examples. It does not amend SAFER Protocol v1.1 or establish new conformance requirements.

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Reference note: Verify project requirements against the adopted NFPA 72 edition, the AHJ, and authoritative manufacturer documentation. Other cited materials support the architectural and historical comparisons in this paper.